Here are the instructions for checking RoboRio crash details. This is applicable, only if the RoboRio crash was caused by an application fault. If a power failure or loose wiring caused a RoboRio crash, there is nothing we can do to investigate further.

RoboRio runs a version of the Linux operating system. Hence, we can connect to it directly to review the logs.

- Ensure that you are connected to the Robot via Wi-Fi or Ethernet cable.
- Find and run "Putty" on the driver station laptop. In the window that opens, double click the "Robot" entry - this will establish a connection with the Robot.
- When prompted, enter the user id as "lvuser". This user has no password.
- Once connected, navigate to the /tmp folder and look for log files. RoboRio creates all temporary files in the /tmp folder. Unfortunately, Rio creates these files in its RAM. So, anytime the Rio is rebooted, this folder is cleared. If you do not find any .log files in this folder, then it means that the Rio was completely rebooted.
- Run these commands:
  - cd /tmp
  - ls -al
- This will list all the files. Once you find a log file that you would like to view, run this command:
  - cat log_file_name.log
- This will display the contents of the log file on the screen. Copy the content from the screen and paste into notepad on the local machine.
- Close the Putty window.

Here's a sample of a log file and where to look to identify the issue. You will notice below that the Java Runtime Environment (JRE) has crashed. Once this crashes, the whole JRE has to reboot and then the application has to be started again. The lines highlighted in red identifies the problem area – in this case, OpenCV has crashed. OpenCV is what we use for video processing.

```
#
# A fatal error has been detected by the Java Runtime Environment:
#
#  SIGBUS (0x7) at pc=0xad323b9a, pid=1429, tid=2892047456
#
# JRE version: Java(TM) SE Embedded Runtime Environment (8.0_06-b23) (build 1.8.0_06-
b23)
# Java VM: Java HotSpot(TM) Embedded Client VM (25.6-b23 mixed mode linux-arm )
# Problematic frame:
# C  [libopencv_core.so.3.1+0x14bb9a]  cv::_OutputArray::create(int, int const*, int,
int, bool, int) const+0x8b1
#
# Core dump written. Default location: //core or core.1429 (max size 2048 kB). To
ensure a full core dump, try "ulimit -c unlimited" before starting Java again
#
# If you would like to submit a bug report, please visit:
```

```
#   http://bugreport.sun.com/bugreport/crash.jsp
# The crash happened outside the Java Virtual Machine in native code.
# See problematic frame for where to report the bug.
#

---------------  T H R E A D  ---------------

Current thread (0xacd68000):  JavaThread "Thread-0" daemon [_thread_in_native,
id=1468, stack(0xac5c3000,0xac613000)]
```